Identifying Individualized Multiple Chronic Condition (MCC) Patterns Associated with Dementia

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Background

Dementia is one of the leading causes of death among the older population that impairs one's cognitive function, memory, and ability to perform daily activities. Managing dementia patients is challenging because

- Patients have more difficulty expressing their discomfort
- Clinical presentations may be very different
- Dementia patients are 2.7 times more likely to have two or more existing chronic conditions – multiple chronic conditions (MCC)

Studies have shown that patients with MCC are more susceptible to developing dementia. Although current literature has studied the common MCC patterns that might lead to dementia, most of them focused on cluster analysis rather than MCC interactions. Moreover, the majority also focused on population-level data, which might not accurately reflect an individual's risks. Knowing which MCC interactions might worsen cognitive decline when coupled with the patient's existing MCC profiles may help delay or prevent the developing ADRD by preventing these particular MCC and interactions.

Objective

To fill this gap, we proposed a proof-of-concept study to identify personalized combinations of multimorbidity (or MCC patterns) that likely (or less likely) result in dementia when coupling them with patients existing comorbidities for each individual.

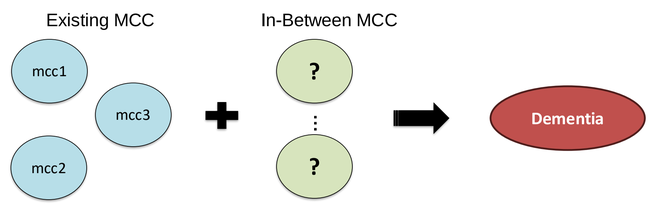

Fig 1: Research Question.

Methods

Data Source

This study used survey data called Uniform Data Set from one of the largest research databases for Alzheimer's disease in the U.S., the National Alzheimer's Coordinating Center (NACC).

Acknowledgment

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Study Design

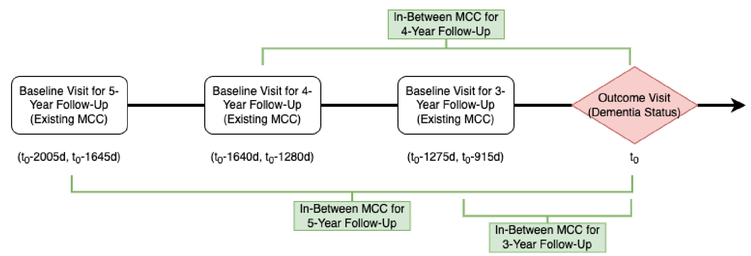

Fig 2: Study Timeline. We first identified the first time the patient was diagnosed with dementia or their last visit to NACC (outcome visit). Then, using the outcome visit as the anchor, the baseline visit was defined as the patient visit that happened 3, 4, and 5 years, respectively, before the outcome visit.

Multiple Chronic Conditions (MCC)

We identified 16 MCC in total aside from Dementia. For each patient, we recorded their existing MCC at baseline visits.

Model Setup

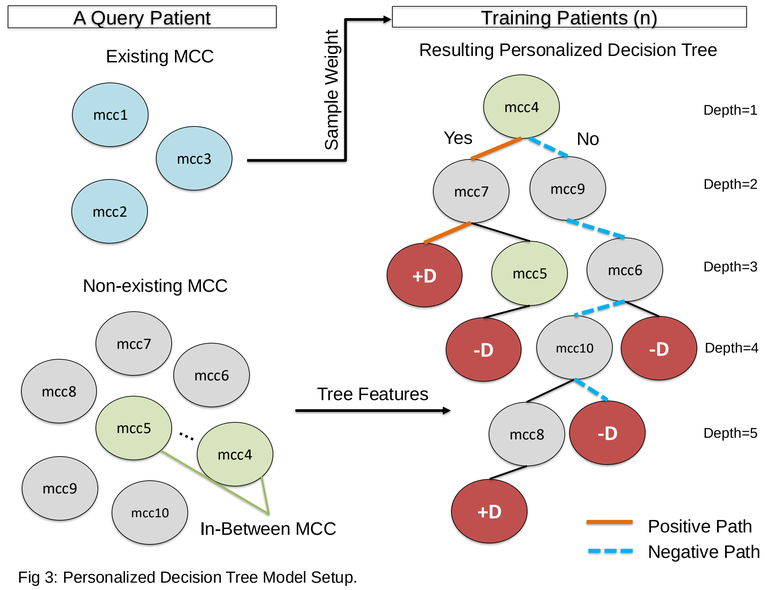

Fig 3: Personalized Decision Tree Model Setup.

lazyDT

Sample Weighting: 4.5^d for every disease (d) that the training patient had in common with the query patient's existing MCC

Decision Tree Model: Top-down decision tree with maximum depth=5, cost complexity pruning=0.001 with adjusted class weight.

eagerDT

Sample Weighting: 1 if the training patient had all the query patient's existing MCC (included) else 0 (excluded)

Decision Tree Model: same as lazyDT

randomDT

Using the structure of lazyDT, we introduced randomness by randomly selecting an MCC to be the split node for every split nodes in lazyDT.

Results

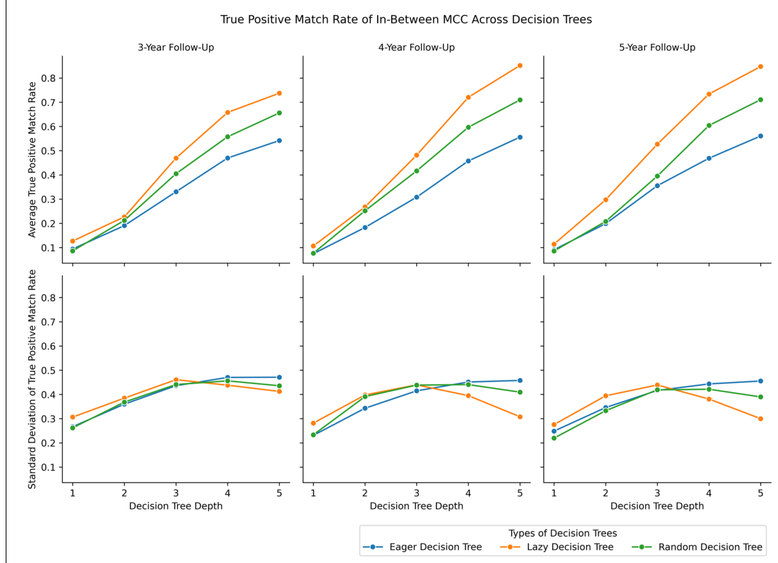

Fig 3: True Positive Match Rate of In-Between MCC.

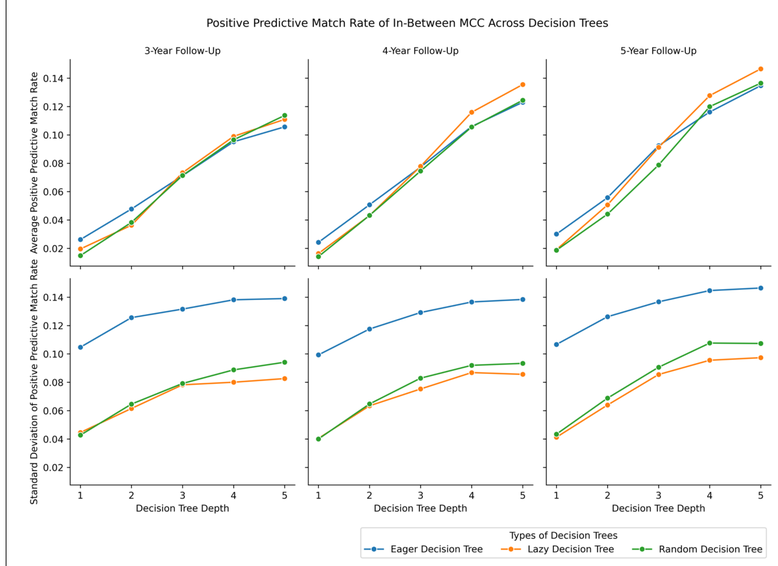

Fig 4: Positive Predictive Match Rate of In-Between MCC.

Conclusion

Our study demonstrated the use of decision tree models to predict future dementia status and potential high-dementia-risk MCC patterns (in-between MCC) by examining important features at split nodes. Our results showed that lazyDT had better predictive performance and a better ability to identify potential MCC patterns that may or may not end up with ADRD than eagerDT. Future directions include incorporating syndemic factors into individual profiles, consolidating these MCC interactions in groups, and identifying the degree of association between MCC interactions and dementia.

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